

Module 07: Communication — Interdisciplinary Methods and Collaborative Research

Course: Active Inference 401 | **Unit:** Research Methods | **Audience:** Graduate students / researchers

Learning Objectives

1. Design **interdisciplinary research** programs that apply Active Inference across domains.
2. Analyze methods for **communicating Active Inference** to non-specialist audiences.
3. Evaluate the role of **open science, reproducibility, and community building** in advancing the field.

Key Concepts

1. Interdisciplinary Research Design

Active Inference is inherently interdisciplinary, spanning neuroscience, psychology, philosophy, physics, mathematics, robotics, and ecology. This creates both opportunities and challenges:

Bridging disciplines: Successful interdisciplinary projects require:

- A shared formal language (mathematics of free energy provides this)
- Translation protocols between domain-specific vocabularies
- Mutual respect for disciplinary expertise and methodological standards
- Clear specification of what each discipline contributes

Cross-domain application areas:

Domain	Active Inference Application	Methods Used
Neuroscience	Brain as inference engine	DCM, EEG, fMRI (Modules 01, 03)
Psychology	Decision-making, perception	Behavioral paradigms (Module 04)
Psychiatry	Computational disorders	Clinical translation (Module 06)
Robotics	Embodied agents	Implementation (Module 05)
Ecology	Organisms as FEP systems	Bayesian mechanics, field studies
Social science	Collective behavior	Multi-agent models, survey data
Philosophy	Mind, consciousness	Conceptual analysis, thought experiments
AI / ML	Intelligent agents	Algorithm development, benchmarking

2. Communication Strategies

Communicating Active Inference to different audiences requires different approaches:

To neuroscientists: Emphasize empirical predictions, DCM results, neural implementation
To psychologists: Emphasize behavioral predictions, computational phenotyping, clinical applications
To physicists: Emphasize Bayesian mechanics, FEP derivation, connections to thermodynamics
To engineers: Emphasize control theory connections, robot implementations, algorithm design
To clinicians: Emphasize clinical utility, patient outcomes, treatment implications
To the public: Use metaphors (brain as scientist, prediction machine), avoid jargon, emphasize real-world applications

Common misunderstandings to address:

1. "The brain literally calculates Bayes' theorem" → Active Inference is a normative/computational-level theory; neurons need not literally perform Bayesian computation
2. "Free energy principle is unfalsifiable" → Specific implementations make testable predictions; the principle itself is a framework, not a hypothesis

3. "Active Inference solves everything" → It provides a framework; specific applications must be empirically validated

3. Open Science and Reproducibility

Active Inference research benefits from and should contribute to open science:

Open-source software: pymdp, SPM, RxInfer.jl — all publicly available. New tools should be open-source by default.

Pre-registration: Register hypotheses, models, and analysis plans before data collection. This prevents post-hoc model selection bias.

Data sharing: Share behavioral data, fitted parameters, and model specifications. This enables replication and meta-analysis.

Reproducible analysis: Use version-controlled code, containerized environments, and literate programming (Jupyter notebooks, R Markdown) to ensure analyses are reproducible.

4. Community and Collaboration

Active Inference Institute: Central hub for education, research coordination, and community building.

Textbook groups: Reading groups working through Parr, Pezzulo & Friston (2022) and related texts.

Cross-institutional collaborations: Multi-site studies using standardized tasks and models enable larger sample sizes and greater generalizability.

Student mentorship: Graduate students in Active Inference need training in multiple disciplines — mathematics, neuroscience, programming, and philosophy. Mentorship programs should span these areas.

5. Publication and Peer Review

Writing for different journals: Active Inference papers must be adapted for the audience:

- *Neuron, Nature Neuroscience*: Emphasize novel empirical findings
- *PLoS Computational Biology*: Emphasize model development and validation
- *Psychological Review*: Emphasize theoretical contributions
- *Biological Psychiatry*: Emphasize clinical relevance
- *IEEE/ICRA*: Emphasize engineering applications

Review challenges: Papers may be reviewed by experts in one domain who lack expertise in another aspect of the work. Include clear, self-contained introductions to each disciplinary component.

Summary

Interdisciplinary Active Inference research requires bridging domain-specific vocabularies with shared mathematical formalism, adapting communication strategies for different audiences, and embracing open science practices. Community building through the Active Inference Institute and cross-institutional collaboration accelerates the field's development.

Further Reading

- Parr, T., Pezzulo, G. & Friston, K. J. (2022). *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press.
- Active Inference Institute. <https://www.activeinference.org/>
- Ramstead, M. J. D. et al. (2020). An introduction to the free energy principle and related research. *Physics of Life Reviews*, 34, 1-5.